DECK 11 OF 13 · FRC JAVA CRASH COURSE

PathPlanner step by step

Set it up, draw an auto, attach commands, and run it

WPILib 2026 · Java · Command-based

OVERVIEW

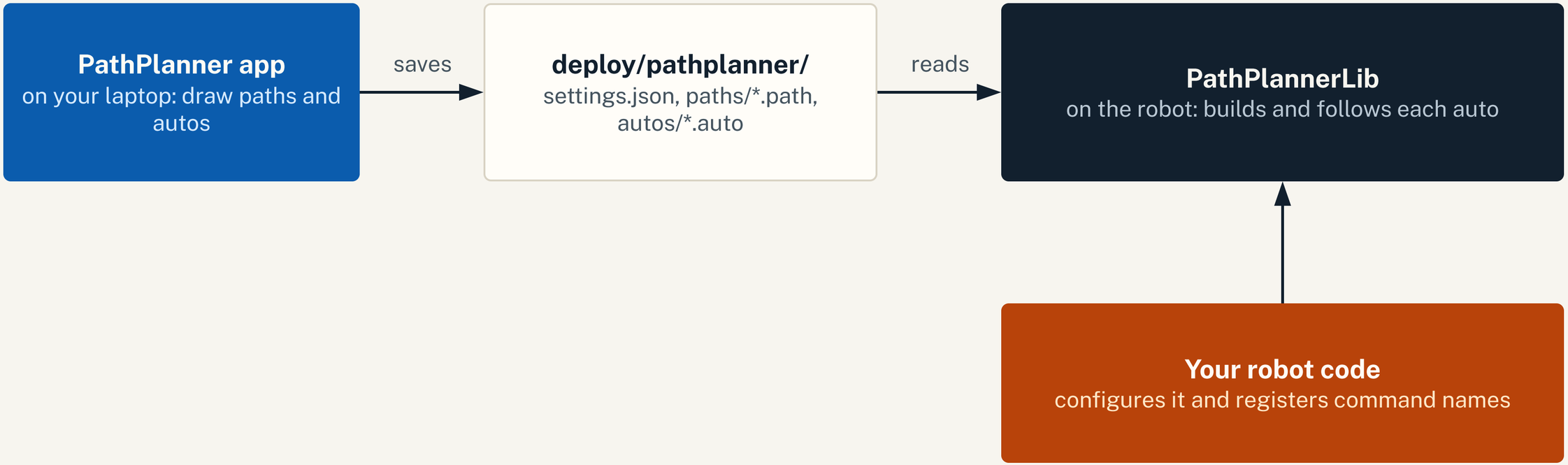
What this deck covers

PathPlanner is two pieces: an app where you draw autos, and a library that runs them on the robot.

- Installing both, and the settings a tank drive needs.
- Drawing paths, and building them into an auto.
- Attaching commands to paths, four different ways.
- The robot code, and a test in the simulator.

HOW IT FITS TOGETHER

The app writes files, and the library runs them



Schematic. The files travel to the roboRIO when you deploy, like the rest of src/main/deploy.

WHAT YOU NEED FIRST

Four methods the drivetrain must already have

Method	What it does	Needs
getPose()	Where the robot thinks it is on the field	Encoders, a gyro, odometry
resetPose(pose)	Tells odometry where the robot is	Odometry
getSpeeds()	How fast the robot is moving, robot-relative	Encoders
driveSpeeds(speeds)	Drives at a forward speed and turn rate	Wheel speed control

Deck 10’s CANDriveSubsystem has all four. This deck reuses it. A stock KitBot has no encoders or gyro.

INSTALL

Install the app and the library

1. **Install the PathPlanner app**
   From the Microsoft Store for automatic updates, or from the GitHub releases page.

2. **Open WPILib VS Code**
   Open your robot project.

3. **Add PathPlannerLib**
   WPILib logo in the activity bar, then find PathplannerLib under Available Dependencies and click Install.

4. **Build once**
   Run WPILib: Build Robot Code to prove the library is in.



PathPlannerLib install

pathplanner.dev

INSTALL

PathPlannerLib in the Dependency Manager

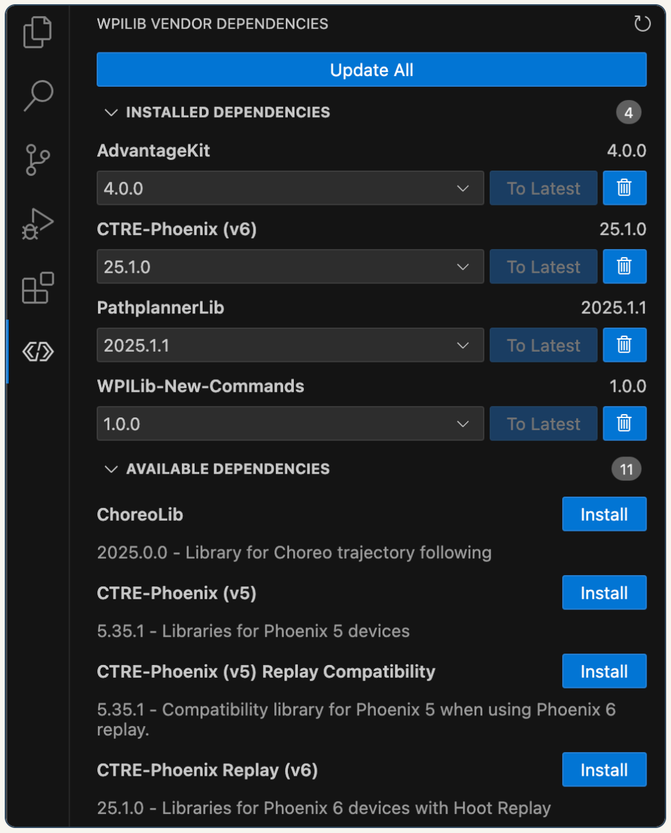


Image: PathPlanner docs, MIT license. The versions shown are from 2025.

Installed Dependencies
PathplannerLib appears here once it is installed.

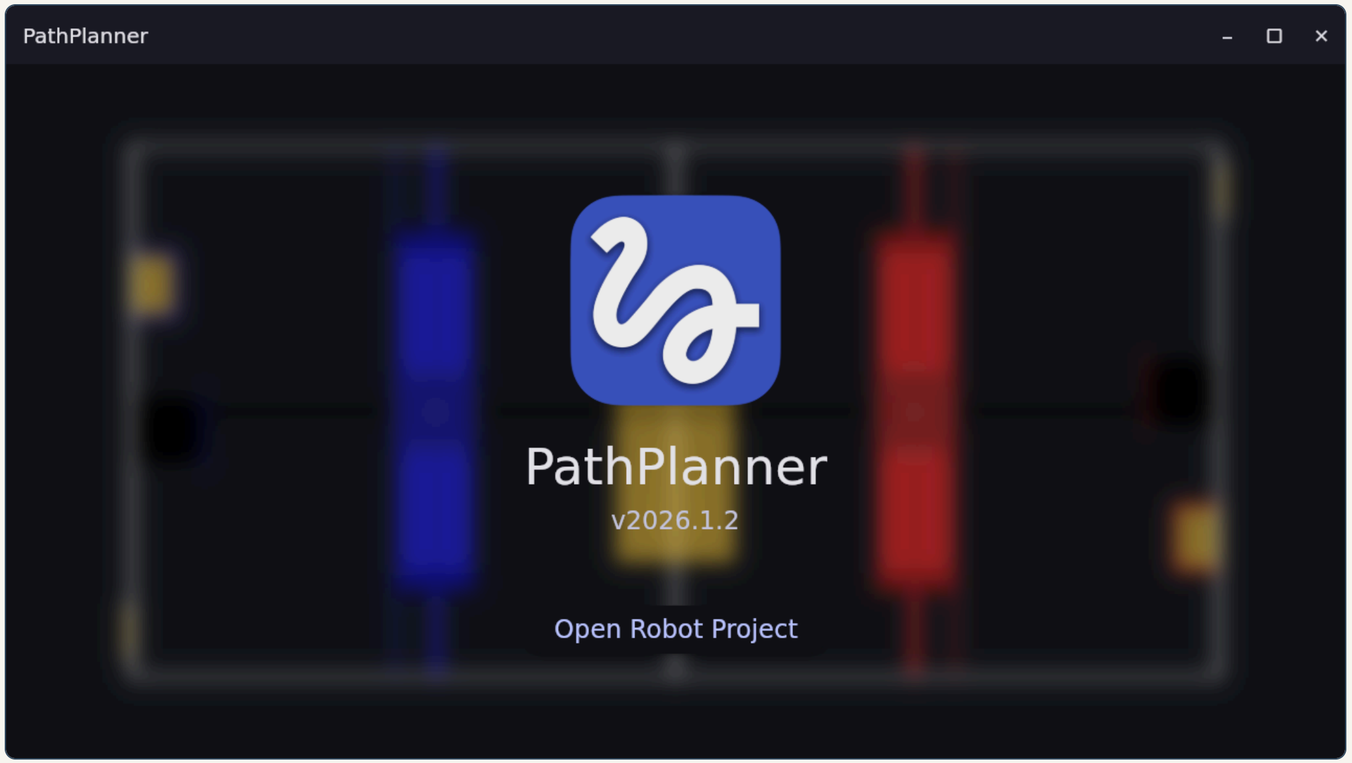
Available Dependencies
Install it from this list the first time.

Update All
Moves every library to its newest version for the season.

PathPlanner docs: PathPlannerLib: Getting Started

THE APP

Open your robot project



Screenshot: PathPlanner 2026.1.2, with this deck’s KitBot settings.

Open Robot Project
Pick the folder that holds build.gradle.

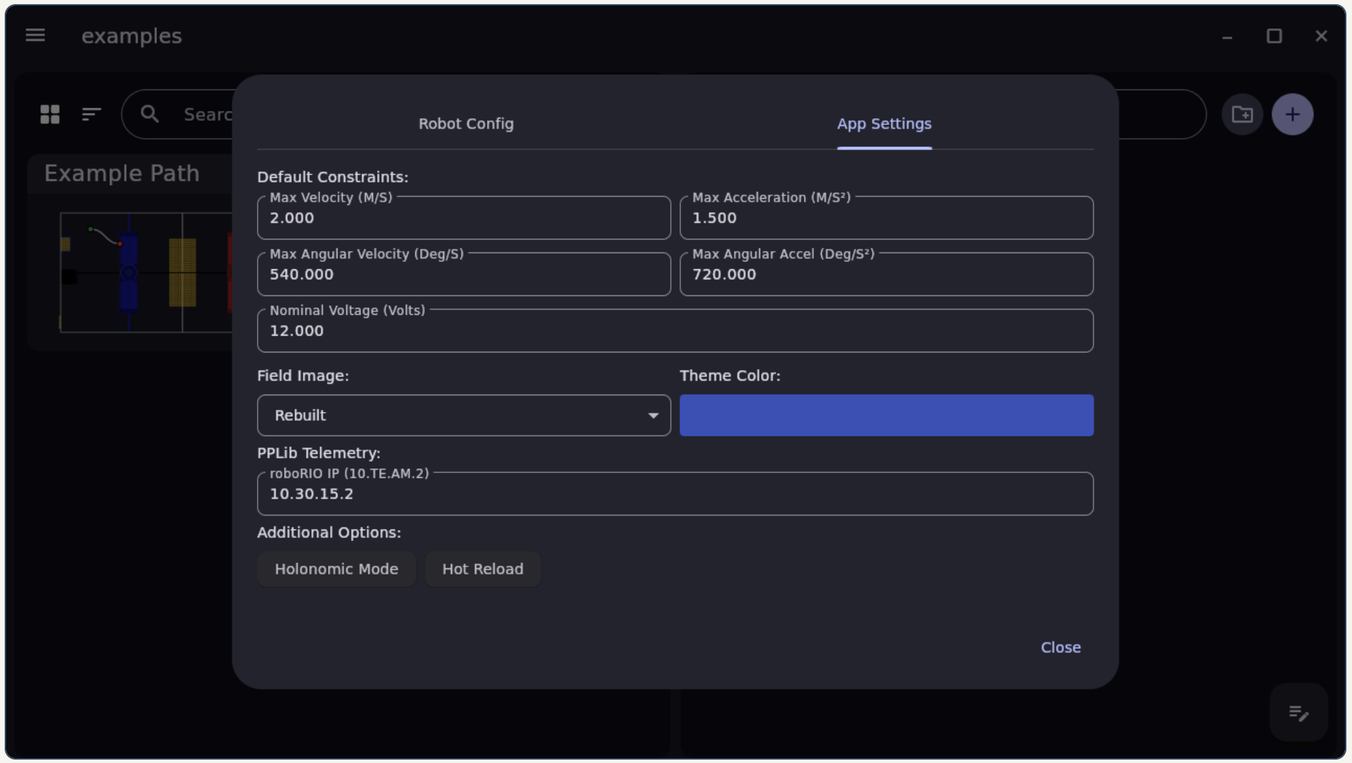
Where files go
The app creates src/main/deploy/pathplanner inside that folder.

The 2026 field
If it offers the 2026 field image, choose Yes.

PathPlanner docs: Getting Started

SETTINGS, PART 1

App settings: tank mode and safe default limits



Screenshot: PathPlanner 2026.1.2, with this deck's KitBot settings.

Holonomic Mode off
A tank drive cannot move sideways. Leave it on only for swerve.

Default constraints
2 m/s and 1.5 m/s^2 for new paths: slow on purpose.

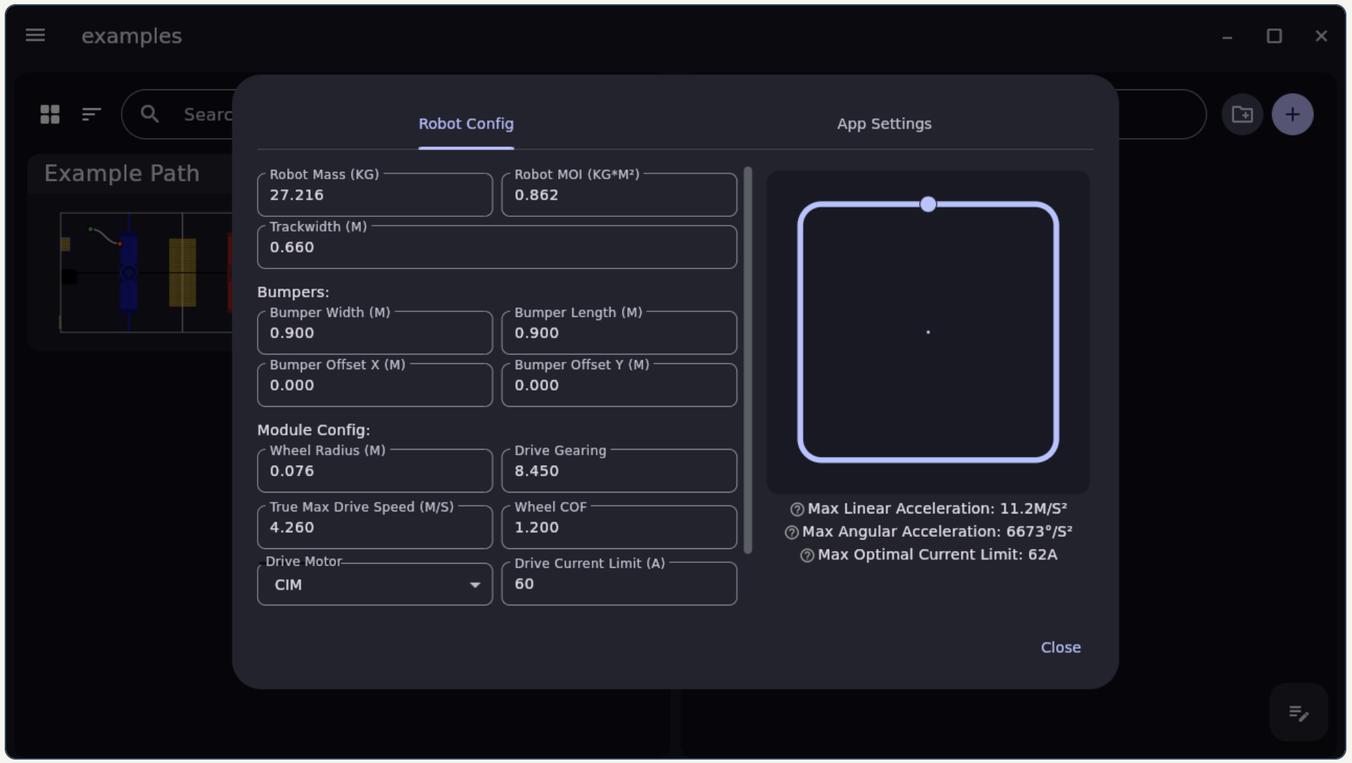
Field image
Rebuilt is the 2026 field.

Hot Reload
Keep it off at events.

PathPlanner docs: Settings

SETTINGS, PART 2

Robot config: the numbers that limit every path



Screenshot: PathPlanner 2026.1.2, with this deck's KitBot settings.

Mass: 27.216 kg, which is 60 lb

MOI: 0.862 kg·m^2, WPILib's KitBot estimate

Trackwidth: 0.66 m, which is 26 in

Wheels: 0.0762 m radius, 8.45:1

True max speed: 4.26 m/s, 85% of free speed

Motor: CIM, 60 A limit

Bumpers: for the drawing only

PathPlanner docs: Robot Config

CONTROL THEORY

Where “Max Linear Acceleration: 11.2” comes from

1. **Grip: 160 N per side**
   Wheel friction × weight on that side: 1.2 × 27.216 kg × 9.8 ÷ 2.

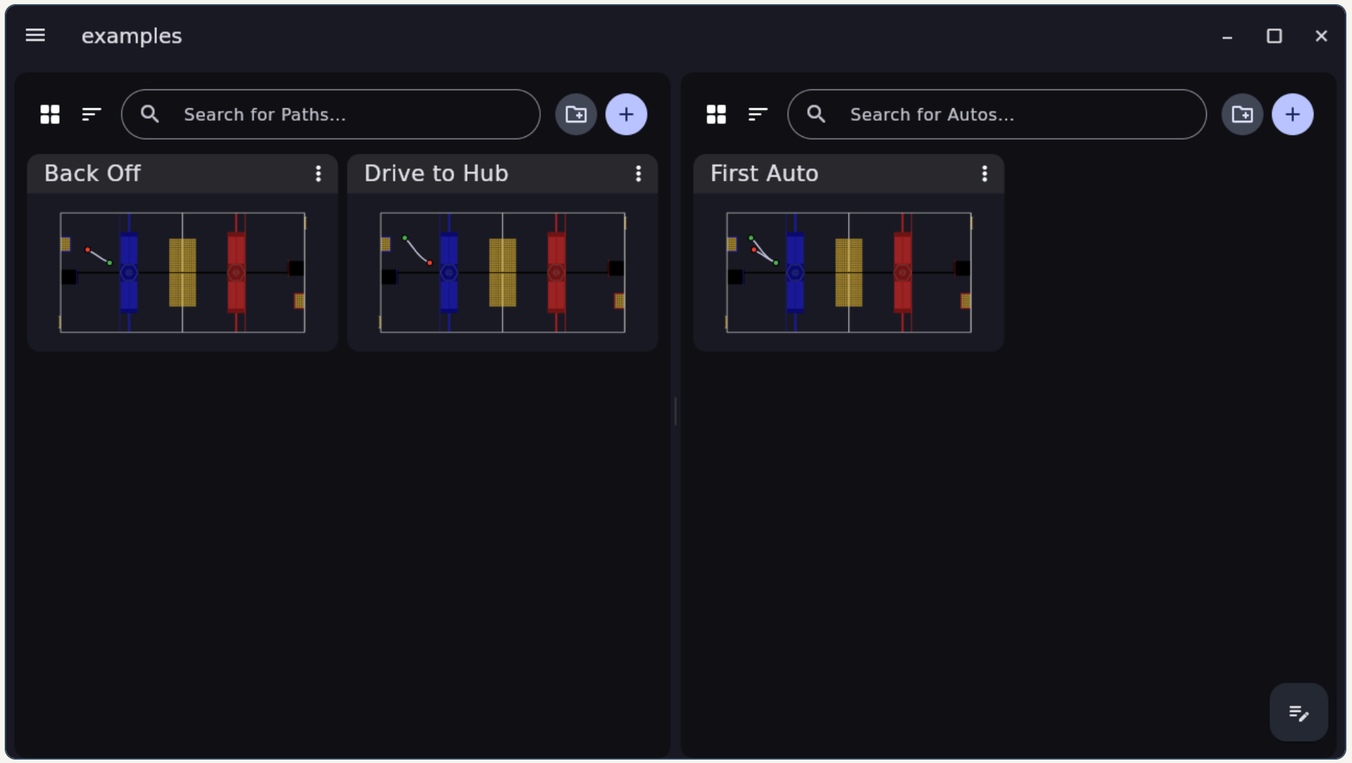
2. **Motors: 152 N per side**
   Two CIMs at 60 A give 18.5 N·m at the wheel. 6.9 N·m goes to friction, judged from true max speed. Divide by the wheel radius.

3. **The smaller one wins: 152 N**
   Two sides push 304 N on 27.216 kg: 11.2 m/s^2.

4. **Slip above 62 A**
   The app’s Max Optimal Current Limit: past it, the wheels slip instead of pushing harder.

THE APP

Paths on the left, autos on the right



Screenshot: PathPlanner 2026.1.2, with this deck's KitBot settings.

Path
One continuous motion from a start to an end.

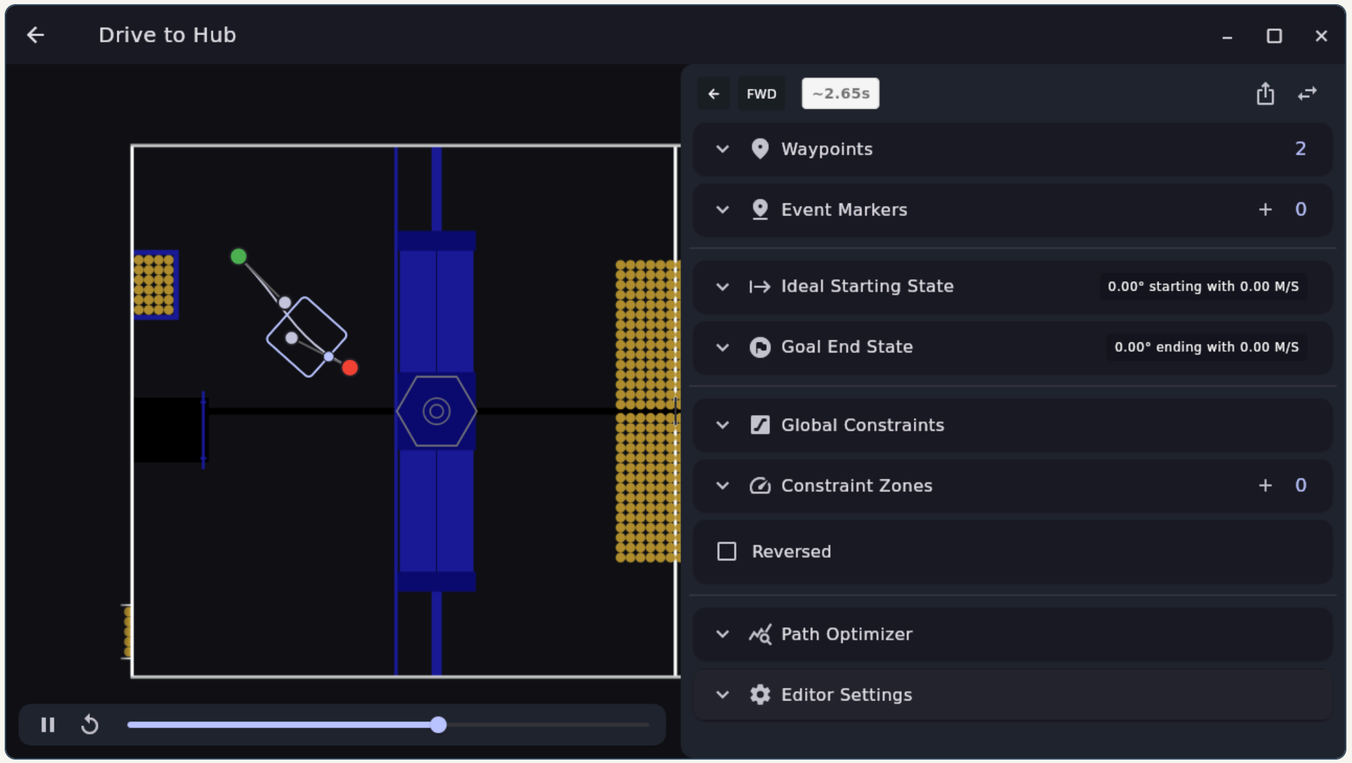
Auto
A whole routine: paths and commands in order.

The + buttons
Make a new path or auto. Click a name to rename it.

The ⋮ menu
Duplicate or delete.

DRAWING A PATH

Drive to Hub: two waypoints and a curve



Screenshot: PathPlanner 2026.1.2, with this deck's KitBot settings.

Waypoints
Green is the start, red the end. Double-click the field to add one.

Heading
The direction of travel through a point.

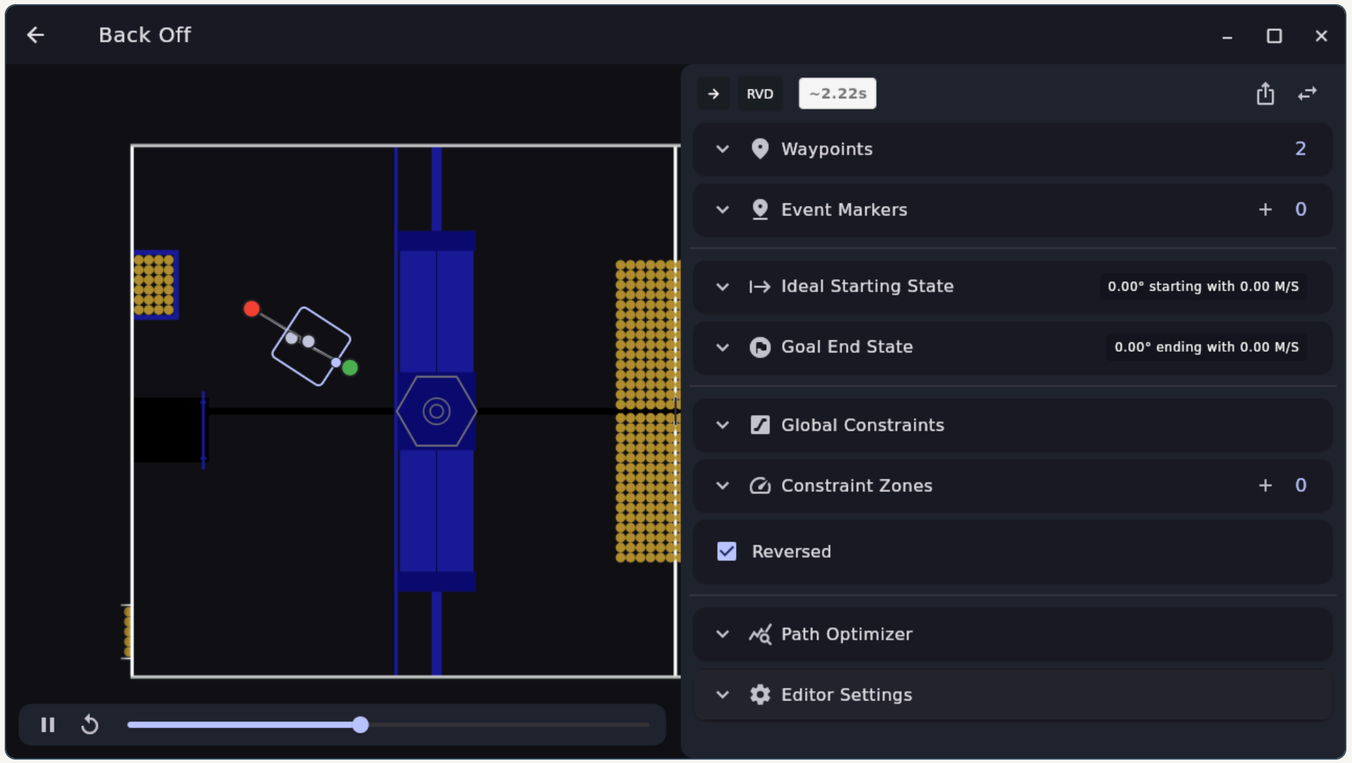
Control points
The grey dots. Drag them to bend the curve.

Time estimate
About 2.65 s with these limits.

PathPlanner docs: Editing Paths and Autos

DRAWING A PATH

Back Off: the same robot, driving backward



Screenshot: PathPlanner 2026.1.2, with this deck’s KitBot settings.

Starts where the last path ended
3.3 m by 4.7 m, so the robot needs no jump.

Reversed
Tick it to drive backward. The panel then shows RVD.

Tank only
Reversed appears with Holonomic Mode off.

PART 1

Commands and paths

Four ways to make the robot do something along the way

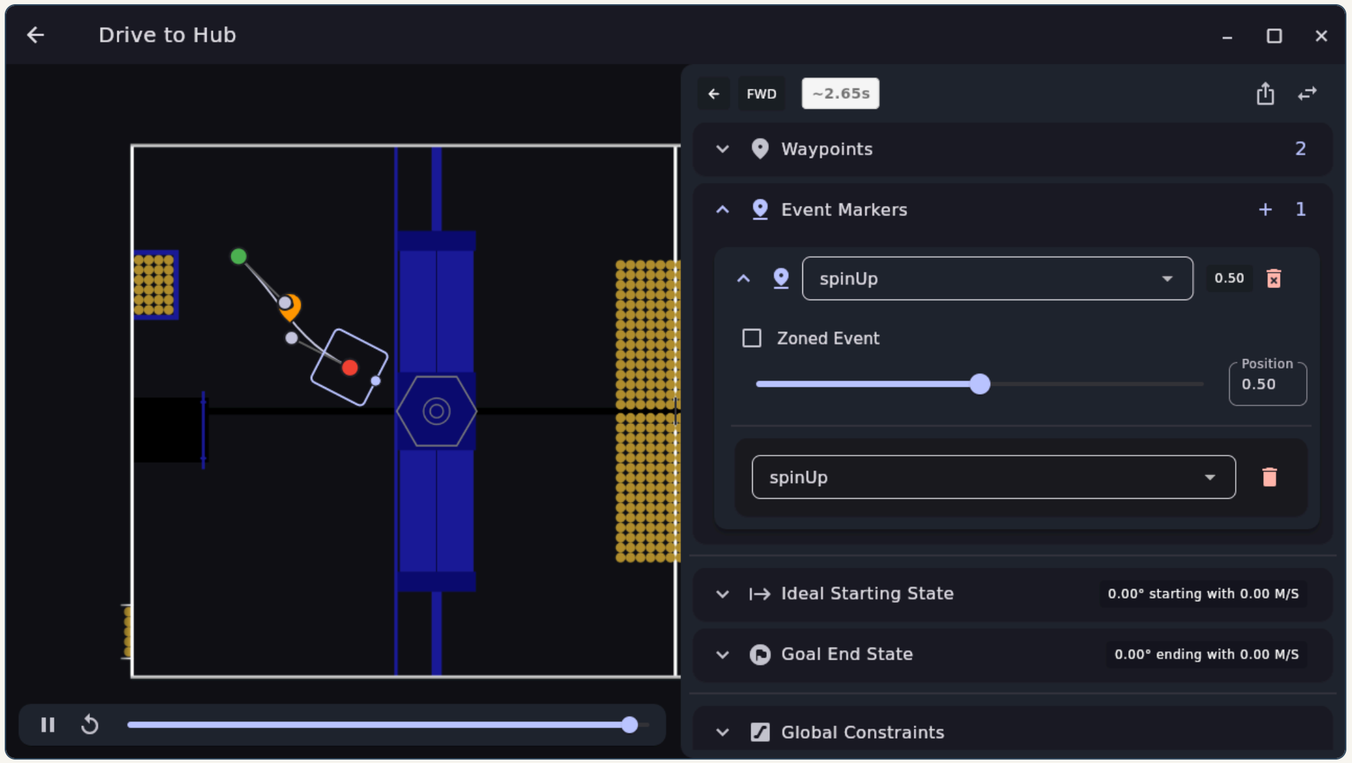
COMMANDS AND PATHS

Four ways to run a command with a path

Where	When it runs	Set up in	In this deck
Named command in an auto	Between paths. The next step waits for it.	The app	launch
Event marker with a command	At a point on a path, while it keeps driving	The app	spinUp
Event trigger	When the robot passes a marker by name	Java	EventTrigger
Java composition	Anywhere, with Deck 2’s andThen and friends	Java	firstAutoInCode

All four end up as ordinary WPILib commands, from the same factory methods.

An event marker runs spinUp while the robot drives



Screenshot: PathPlanner 2026.1.2, with this deck's KitBot settings.

Marker name
spinUp: also the name an event trigger would use.

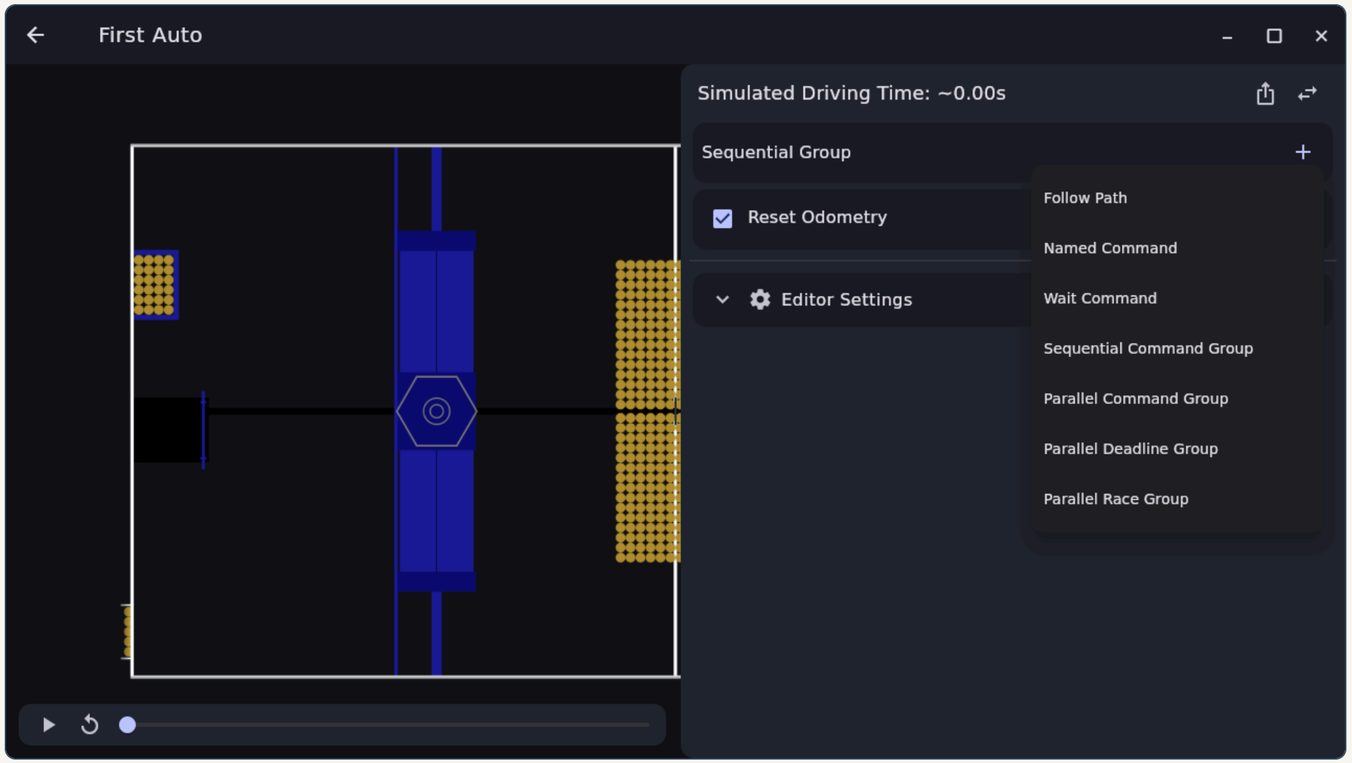
Position 0.50
Halfway between waypoint 0 and waypoint 1.

The + button
Named Command, then type spinUp and press Enter.

Zoned Event
A start and an end position instead of one point.

BUILDING AN AUTO

What an auto can hold



Screenshot: PathPlanner 2026.1.2, with this deck's KitBot settings.

Follow Path
Drive one of your paths.

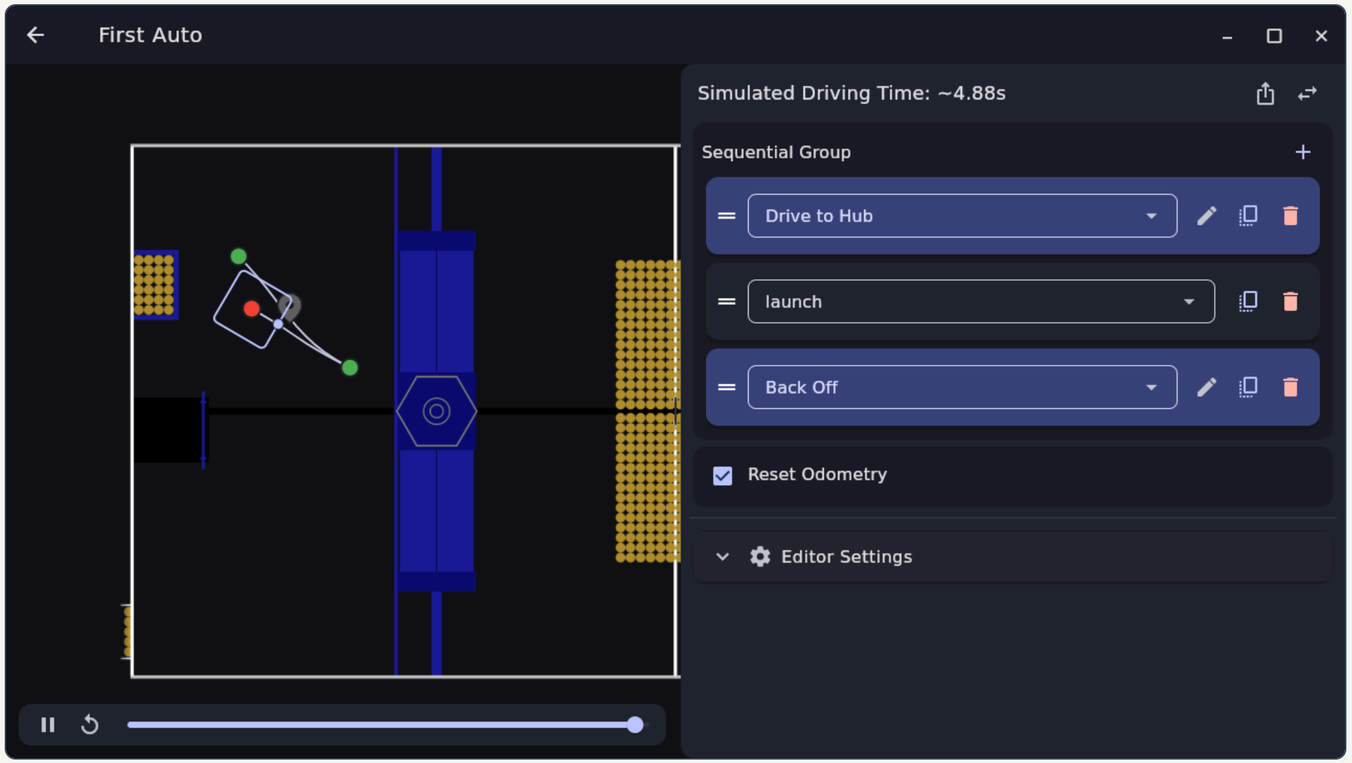
Named Command
Run a command your code registered.

Wait Command
Pause for a set time.

Groups
Sequential, parallel, deadline and race, as in Deck 2.

BUILDING AN AUTO

First Auto: drive, launch, back off



Screenshot: PathPlanner 2026.1.2, with this deck’s KitBot settings.

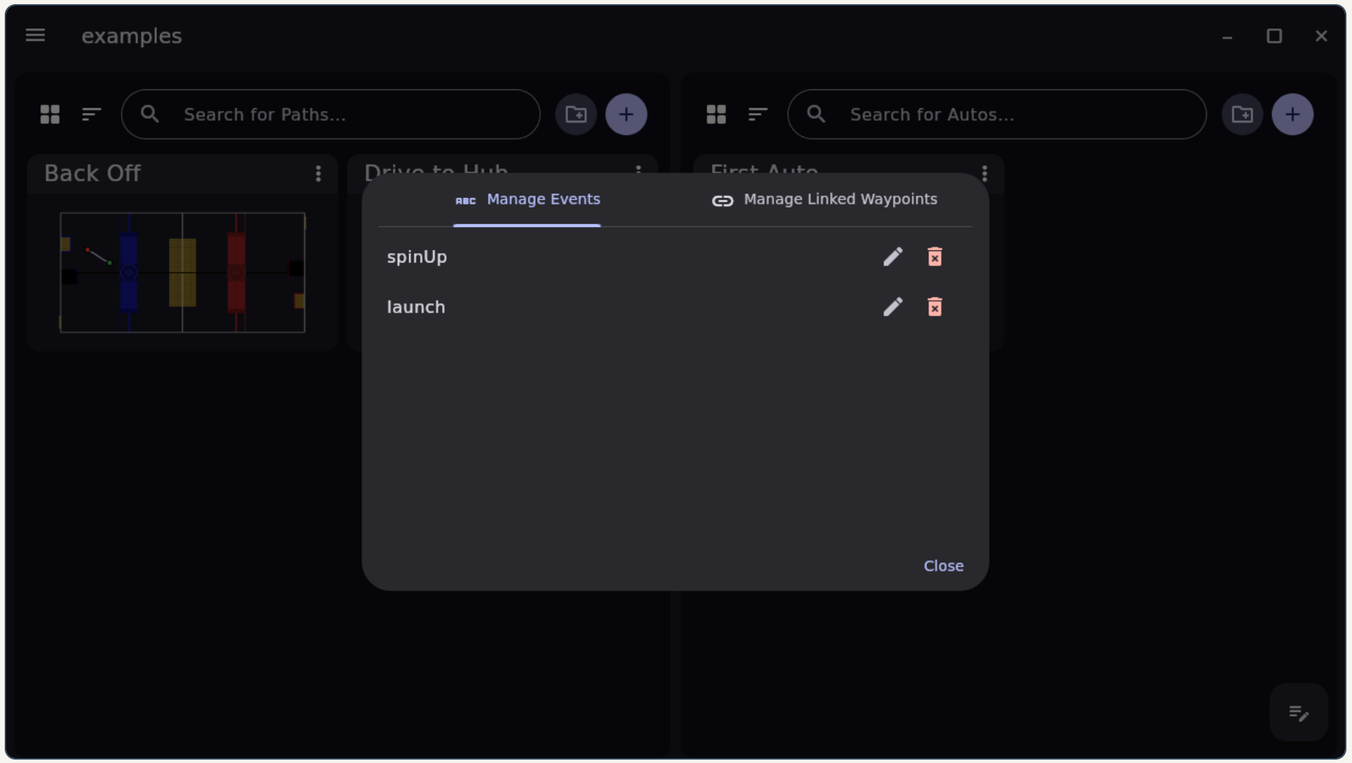
Drive to Hub
Its spinUp marker fires on the way.

launch
A named command. Back Off waits for it.

Back Off
Reversed, away from the hub.

Reset Odometry
Ticked: the robot starts at the first path’s start.

Every name the code must register



Screenshot: PathPlanner 2026.1.2, with this deck’s KitBot settings.

Manage Events
The button at the bottom right of the project browser.

spinUp and launch
Every command name used in any path or auto.

Rename here
Changes the name everywhere it is used.

PART 2

The robot code

Deck 10’s drivetrain, plus a stand-in launcher

THE COMMANDS

The two commands the names point to

Launcher.java

```java
/** Starts the flywheel, then ends at once. The flywheel keeps spinning. */
public Command spinUp() {
  return runOnce(() -> flywheel.set(FLYWHEEL_SPEED)).withName("Spin up");
}

/** Feeds for LAUNCH_SECONDS, then stops both motors. */
public Command launch() {
  return run(() -> feeder.set(FEED_SPEED))
      .withTimeout(LAUNCH_SECONDS)
      .finallyDo(this::stop)
      .withName("Launch");
}
```

Ordinary factories from Deck 2. PathPlanner only decides when they run.

PATHPLANNERLIB

Load the settings the app saved

PathPlannerSetup.java

```
RobotConfig config;
try {
  config = RobotConfig.fromGUISettings(); // deploy/pathplanner/settings.json
} catch (Exception e) {
  DriverStation.reportError("PathPlanner settings: " + e.getMessage(), false);
  return false;
}
```

fromGUISettings() reads deploy/pathplanner/settings.json: the numbers from the Robot Config slide.

PATHPLANNERLIB

Tell PathPlannerLib how to drive this robot

PathPlannerSetup.java

```
  AutoBuilder.configure(
1     drive::getPose, // where odometry thinks we are
1     drive::resetPose, // used by Reset Odometry
1     drive::getSpeeds, // robot-relative speeds
1     (speeds, feedforwards) -> drive.driveSpeeds(speeds),
2     new PPLTVController(0.02), // tank follower, 20 ms loop
      config,
3     PathPlannerSetup::isRedAlliance, // true: mirror for red
      drive);
```

1. **The four methods**
   From the needs slide.

2. **The tank follower**
   PPLTVController, run every 20 ms.

3. **Draw on blue**
   True on red: paths are mirrored.

ROBOTCONTAINER

Register names, then build the chooser

RobotContainer.java

```
// 1. Tell PathPlannerLib how to drive this robot.
boolean ready = PathPlannerSetup.configure(drive);

// 2. Register every name used in the app, before any auto is built.
NamedCommands.registerCommand("spinUp", launcher.spinUp());
NamedCommands.registerCommand("launch", launcher.launch());

// 3. One chooser entry per .auto file in deploy/pathplanner/autos, plus "None".
autoChooser = ready ? AutoBuilder.buildAutoChooser() : new SendableChooser<>();
if (!ready) {
  autoChooser.setDefaultOption("None", Commands.none());
}
SmartDashboard.putData("Auto Chooser", autoChooser);
```

getAutonomousCommand() returns autoChooser.getSelected().

COMMON MISTAKE

The auto drives, but never launches

What you see	Drive to Hub and Back Off run, the launcher never moves, and the Driver Station shows a PathPlanner warning.
Why it happens	registerCommand ran after buildAutoChooser(), or the spelling differs from the app.
The fix	Register every name first, spelled exactly as in Manage Events.

```
PathPlanner attempted to create a command 'launch' that has not
been registered with NamedCommands.registerCommand
```

The same auto, written in Java

PathPlannerSetup.java

```java
/** First Auto, written in Java: the same two paths and the same launch. */
public static Command firstAutoInCode(Launcher launcher) {
  try {
    PathPlannerPath toHub = PathPlannerPath.fromPathFile("Drive to Hub");
    PathPlannerPath backOff = PathPlannerPath.fromPathFile("Back Off");
    return Commands.sequence(
        AutoBuilder.resetOdom(toHub.getStartingDifferentialPose()),
        AutoBuilder.followPath(toHub), // its spinUp marker still fires
        launcher.launch(),
        AutoBuilder.followPath(backOff));
  } catch (Exception e) {
    DriverStation.reportError("Path file: " + e.getMessage(), false);
    return Commands.none();
  }
}
```

BEST PRACTICE

Treat path and command names as code

> **Pick clear names once. Rename only in the app, then fix the Java in the same commit.**

The code loads paths by file name and commands by name. A rename on one side breaks the other without a build error.

TESTING

Telemetry: the target and the real robot

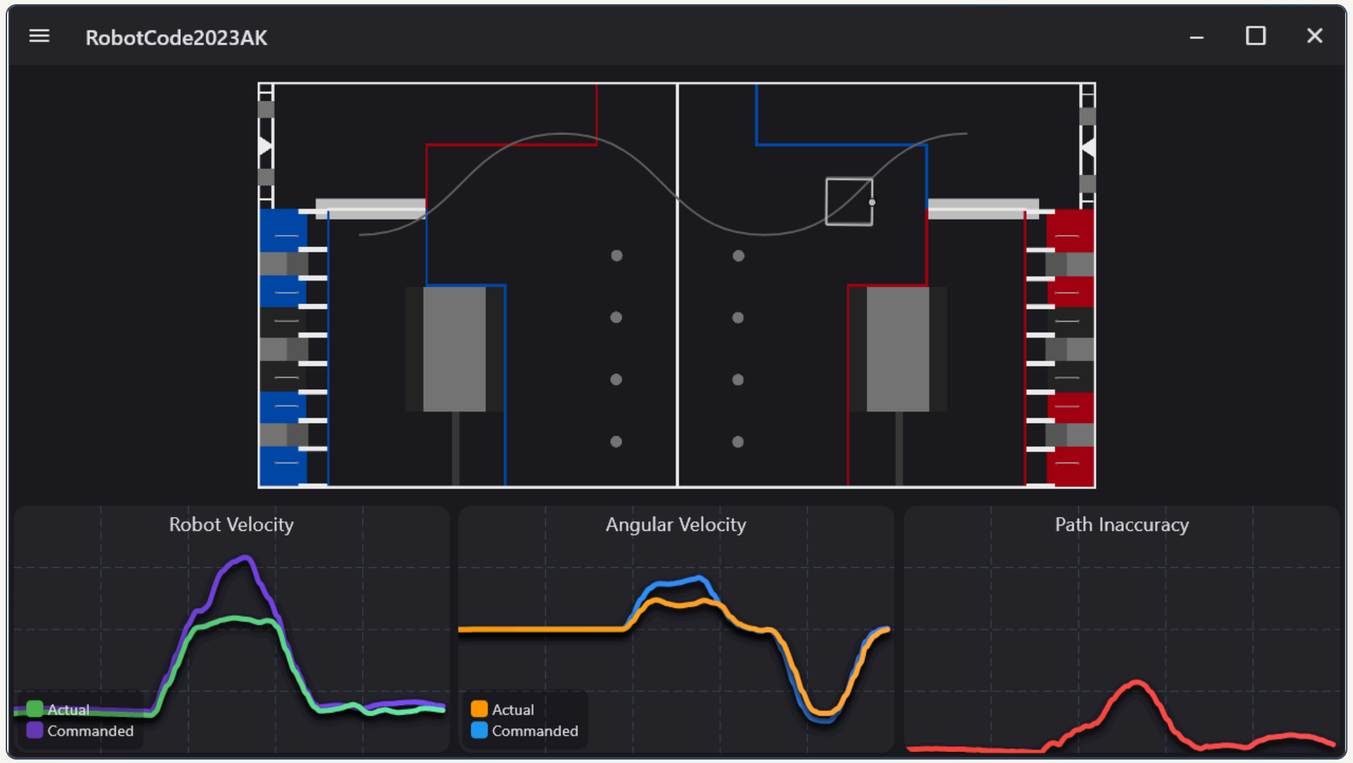


Image: PathPlanner docs, MIT license. An older season's field.

Field preview
The path, the target pose in grey and the real pose in white.

Velocity graphs
Commanded and actual, straight and turning.

Path inaccuracy
How far the robot is from the target, 0 to 1 m.

Simulator or Robot
A switch on the page. Simulator connects to 127.0.0.1.

PathPlanner docs: Telemetry

EXERCISE

Run First Auto in the simulator

1. In a copy of examples, set includeDesktopSupport to true. Make Robot.java create frc.robot.deck11.RobotContainer.

2. Run **WPILib: Simulate Robot Code**. In PathPlanner, open **Telemetry** and choose **Simulator**.

3. In Elastic, set **Auto Chooser** to First Auto. In the Sim GUI, click **Autonomous**.

4. In the app, give Drive to Hub its own max acceleration of 1.0 m/s^2. Restart the simulation and run it again.

Done when: the robot drives to the hub and backs off, with no PathPlanner warnings, and the second run takes longer.

RECAP

What this deck covered

- The app saves files into deploy/pathplanner, and PathPlannerLib runs them.
- Tank drive: Holonomic Mode off, real robot numbers, slow default limits.
- Paths are pieces of motion. An auto strings paths and commands together.
- Commands join paths four ways: auto steps, markers, event triggers and Java.
- Register every name before building autos, spelled exactly as in the app.

Next: Deck 12 covers vision, which corrects the pose every path depends on.

REFERENCE

Official docs



PathPlanner app

pathplanner.dev



Robot config

pathplanner.dev



Build an auto

pathplanner.dev



Named commands

pathplanner.dev